BEHAVIOR Challenge 2026 · Team report

# Harvesting failures for imitation

Human-gated and privileged DAgger in the 2026 BEHAVIOR Challenge: a recovery pipeline for a simulated household robot, the data it produced, and what it changed.

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# Compounding errors and the human gate

## A cloned policy drifts

With error ε per step on the expert's states and T steps per episode, a cloned policy's cost can exceed the expert's by up to $T^2\varepsilon$.

DAgger labels the states the learner itself visits and retrains on all data so far: the bound falls to the order of $T\varepsilon$.

The guarantee needs repeated rounds.

## HG-DAgger: the human holds the gate

The learner drives. The human takes over when the state looks unsafe, keeps full control, and hands back.

Only the takeover segments are labelled.

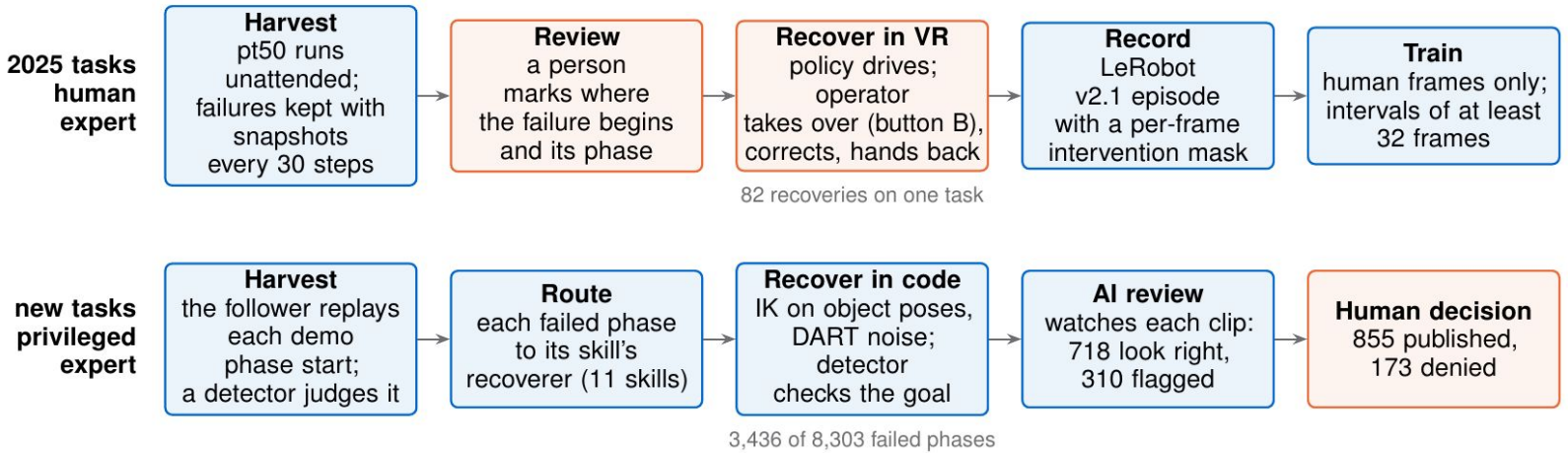
The human is also the bottleneck: every recovery needs a person to record it and to check it.

# The 2026 BEHAVIOR Challenge

| | |
|---|---|
| **Tasks** | 100 household tasks in OmniGibson, 50 of them new; 7 scenes |
| **Robot** | R1Pro: wheeled, two arms, a four-joint torso |
| **Inputs** | RGB and depth from a head and two wrist cameras, plus proprioception; privileged state in training only |
| **Score** | Q-score: the share of a task's goal predicates true at the end, averaged over tasks |
| **Data** | 20,000 demonstrations (200 per task, 1,950 hours), 31 annotated skills |
| **Policies** | pt50, Team Comet's π0.5 for the 2025 tasks; the follower, the team's π0.5 that runs one phase at a time on the new tasks |

Submission deadline 16 October 2026; teams report one rollout on public instances 0 to 9 of each task.

# The recovery pipeline




Top: HG-DAgger with a human expert on the 2025 tasks. Bottom: the scaled variant on the 50 new tasks, where per-skill controllers replace the human expert and a detector replaces the human gate. Orange: people. Blue: the policy, code or an AI agent. Counts as of 14 October 2026.

# Human-gated recovery in VR

1. The policy drives, served over the challenge's websocket protocol.
2. The operator presses B to take over. The arms switch to inverse-kinematics controllers that follow the operator's hands.
3. The operator corrects the robot the way the demonstrations do, and presses B again to hand back.
4. The console records a 15-second tail of policy driving and saves the episode.




Picking up trash: head and wrist cameras; the overlay turns red while the operator drives.

A LeRobot v2.1 episode with a per-frame intervention mask. Quest 3S or Pico 4 headset over CloudXR, 30 Hz.

# Privileged per-skill recoverers

Scripts for 11 skills read the simulator state and solve inverse kinematics on object poses. AI coding agents wrote and iterated them (chop: nine versions).

As in DART, the executed actions are perturbed while the labels stay clean.

A privileged phase detector confirms the goal: 3,436 of 8,303 failed phases recovered (41%).

Seven skills go through review; the rest are published on the detector gate alone.



Chop: halving a Brussels sprout (left wrist, right wrist, head).

The detector judges only the end state. A clip can pass because the goal holds at the end even if the robot got there badly.

# AI review before the human decision

**1,028**

clips reviewed by the agent

**718 / 310**

“looks right” / flagged

**855 / 173**

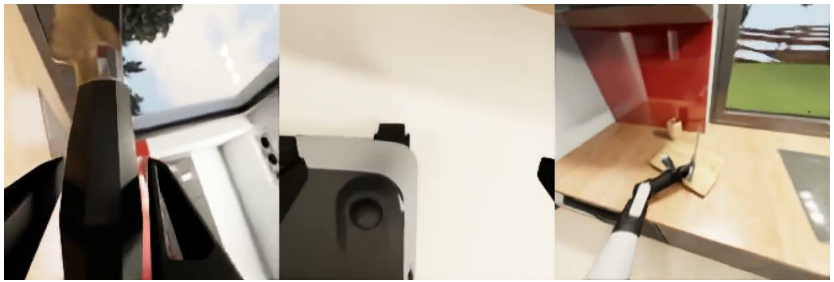
published / denied by a person

**92%**

of the denials had been flagged

An agent running Claude Opus 5.5 watches each clip and writes one line on how the robot reached the goal (“Fingertip on burner knob marker, red→green, pan untouched”). A person then accepts or denies it.

The person accepted 98% of the clips the agent called right and denied 51% of the flagged ones: a cautious filter.



Denied: the detector passed this chop, but the egg was cut by accident.

# The strongest 2025 policy as the expert

**4,181**

attempts from the follower’s failed phases, both rounds

**754**

reached the phase goal (18%)

**500**

confirmed corrections on 35 new tasks

**66**

segments on two 2025 tasks, earlier

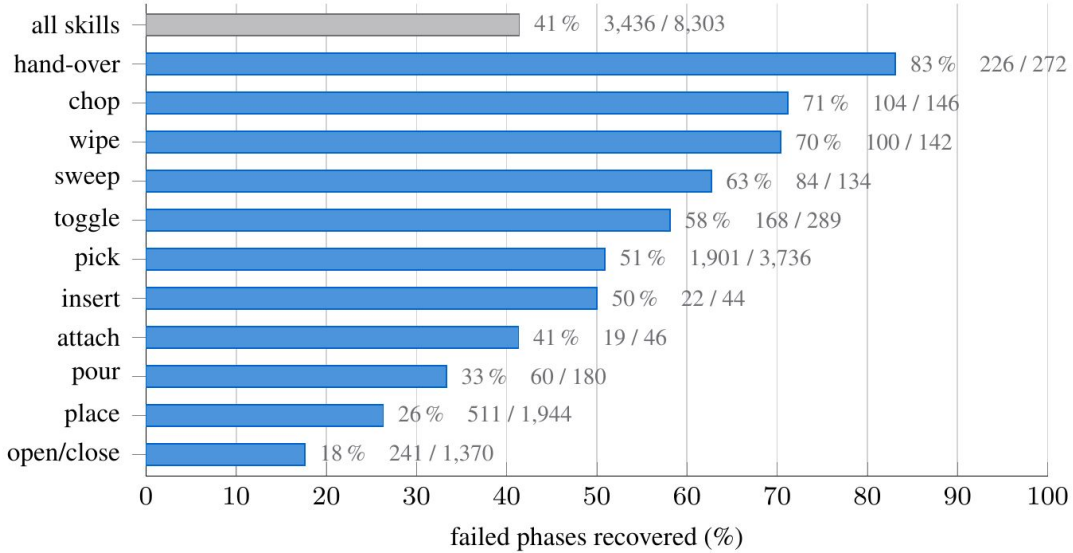
Many objects and skills of the new tasks also appear in the 2025 tasks. For a failed pick-up or place phase, the fleet restores the snapshot and lets pt50 drive, prompted with a 2025 task of the same skill; the detector keeps the attempts that reach the goal.

Unlike the scripts, pt50’s actions come from a policy trained on the same kind of teleoperation as the demonstrations. The 500 corrections of the first round are the recovery data of the submitted follower.

# Recovery data by expert

| Expert and gate | Tasks | Recordings | Used in |
|---|---|---|---|
| **Human in VR, takeover from a rollout (HG-DAgger)** | 2 (2025) | 84 (115 merged episodes) | Pilot 1, not trained |
| **Human in VR, from the failed state** | new | 111 | end-of-phase detector (66) |
| **Privileged per-skill controllers, detector gate** | 48 (new) | 4,166 on 3,739 cases | fine-tunes (hurt); planner |
| **pt50 on the 2025 tasks** | 2 (2025) | 66 segments | instr100_recov, c_gp1k |
| **pt50 on the new tasks, detector gate** | 35 (new) | 500 pick/place, +254 later | submitted follower (d_optB_1000) |

# Recovered share of failed phases by skill




Contact-rich skills with a dedicated recoverer recover best: hand-over 83%, chop 71%, wipe 70%.

The skills where the follower fails most recover worst: place 26% and open/close 18%, which has no dedicated recoverer.

Two harvest batches on the new tasks; review board of 14 October 2026.

# Fine-tuning the follower on recoveries

| Run | Recovery data and schedule | Result and comparator |
|---|---|---|
| **c_gp1k** | 66 pt50 segments of two 2025 tasks, 1,000 steps | held-out success 44% → 48.7% (team report, not reproduced) |
| **r50** | privileged recoveries, 10,000 steps | 46.5% → 11.6% vs the follower, −34.9 [−40.6, −29.6] |
| **d_scr** | privileged, waist twist zeroed, 1,000 steps | mixed phases −0.5 [−8.4, +8.9] vs the control |
| **d_optB_1000 (submitted)** | 500 pt50 corrections of new-task failures, 1,000 steps | pick-ups 37.6% → 44.6%, +7.0 [+1.0, +12.7] vs the control |

Control e_demos1k (1,000 steps, no recovery data); 95% bootstrap intervals over tasks; +7.0 on 383 pick-up phases of 49 tasks. r50's collapse is attributed mainly to the waist joint, not separated from fumbling and the long schedule.

# Recoveries as images for the planner

**84.4%**

phase-classifier accuracy (seconds), from 80.8%, on 198 rollouts

**+3.6 points**

paired over the 48 tasks with recoveries

**73%**

on-time switches on successful attempts (old rule 24%)

**34%**

wrong switches on stuck attempts; old rule 64%

The follower needs a planner that sequences the phase prompts and decides when a phase is done. Each recovery shows the robot in a state the demonstrations never reach, labelled with its phase: it helps as an image even where it hurts as actions.

The submitted end-of-phase detector is trained on the demonstrations, 3,830 follower rollouts and 4,227 recoveries (4,161 privileged, 66 human); its records do not isolate the recoveries' share of its gain.

# Closed loop and the submission

Closed-loop pilot of 16 October, ten new tasks on public instances 0 and 1: no arm completed a task. Arm B, the d_optB_1000 follower with the end-of-phase detector and a skill-scaled timer, scored mean Q 0.060 with 5 of 20 rollouts above zero, and became the submission.

With oracle phase labels, the follower completed no task in 394 rollouts. At this level the follower limits the score, not the phase label.

**0.155**

mean Q of pt50 on the 2025 tasks, 18 successes in 500 rollouts

**0.018**

mean Q on the new tasks at 18:00 HST, 120 rollouts, no success

**0.092 · 1.8%**

leaderboard Q-score and success rate, self-reported, 17 October 2026

Recovery data reaches the submission in the follower and in the end-of-phase detector.

# What limited the method

- **Prevention, not recovery**: operators resumed before the mark, so 29 of 33 place recordings began with the can already in hand. Since 1 October the resume point is 3 s after the mark.
- **Expert time was the bottleneck**: 71% of the harvested failures could not be fixed by a short correction; people recorded 78 recoveries in four days, on one task.
- **A policy's actions, not a script's (untested hypothesis)**: pt50 (18% yield) helped and the privileged recoverers (41%) hurt, perhaps because pt50's corrections stay inside the action distribution the follower learned.
- **A joint no demonstration moves**: the scripts used the waist, and every check passed those clips. Compare per-joint actions with the demonstrations before training.
- **One round, not DAgger**: at most one training round per dataset; the gain is a pick-up phase success under a privileged detector (mixed skills +1.6 [−3.4, +6.8]); no Q-score gain shown.

Next: a second DAgger round with the improved follower; corrections for the other skills; a per-joint check of every recovery before training.